# Bendong Tan

Department of Electrical and Computer Engineering Phone: 19592320448
University of Connecticut Email: bendong.tan@uconn.edu
Storrs, CT, USA Web: https://tbendong.github.io/

## EDUCATION

- **University of Connecticut (Ph.D. candidate in Electrical Engineering)** Jan. 2022 – Now
  - **Research area:** Low-inertia power system identification, monitoring and control
  - **Supervisor:** Prof. Junbo Zhao
- **Mississippi State University (Ph.D. student in Electrical Engineering)** Aug. 2021 – Dec. 2021
  - **Research area:** Low-inertia power system identification, monitoring and control
  - **Supervisor:** Prof. Junbo Zhao
- **Wuhan University (M.S. in Electrical Engineering)** Sept. 2017 – May 2020
  - **Thesis:** Power System Transient Stability Assessment with Unsatisfactory Data
  - **Supervisor:** Prof. Yuanzhang Sun & Prof. Jun Yang
- **Wuhan University (B.E. in Electrical Engineering)** Sept. 2013 – July 2017
  - **Grade Point Average (GPA):** 3.60/4.00
  - **Ranking:** 22/379 **(top 6%)**
  - **Thesis:** Transient Stability Assessment of Power System Based on Data Mining Technology
  - **Supervisor:** Prof. Jun Yang

## WORK EXPERIENCE

- **Teaching Assistant: University of Connecticut, USA** Sept. 2023 –Dec. 2023
  - Responsible for teaching usage of PSS/E, and grading assignments and exams.
- **Internship: National Renewable Energy Laboratory, USA** May 2023 – Aug. 2023
  - Responsible for developing online regional inertia monitoring techniques for WECC power system.

## RESEARCH INTERESTS

Power system dynamics, stability control, estimation, machine learning, renewable energy integration, risk assessment, uncertainty quantification and management.

## PUBLICATIONS

- **Journal Paper**

[J1] Jiahao Liu, Cheng Wang, Junbo Zhao, **Bendong Tan** and Tianshu Bi, "Simplified transient model of DFIG wind turbine for power system frequency dynamics analysis,", ***IEEE Transactions on Power Systems***, 2023.

[J2] **Bendong Tan**, Junbo Zhao, "Debiased uncertainty quantification approach for probabilistic transient stability assessment,", ***IEEE Transactions on Power Systems***, vol. 38, no. 5, pp. 4954-4957, Sept. 2023.

[J3] **Bendong Tan**, Junbo Zhao, *et al.*, "Amortized Bayesian parameter estimation approach for WECC composite load model,", ***IEEE Transactions on Power Systems***, 2023.

[J4] Guozhou Zhang, Junbo Zhao, Weihao Hu, Di Cao, **Bendong Tan**, Qi Huang and Zhe Chen, "A novel data-driven self-tuning SVC additional fractional-order sliding mode controller for voltage stability with wind generations,", ***IEEE Transactions on Power Systems***, vol. 38, no. 6, pp. 5755-5767, Nov. 2023.

[J5] **Bendong Tan**, Junbo Zhao, *et al.*, "Transferable deep kernel emulator for probabilistic load margin assessment with topology changes, uncertain renewable generations and loads,", ***IEEE Transactions on Power Systems***, vol. 38, no. 6, pp. 5740-5754, Nov. 2023.

[J6] **Bendong Tan**, Junbo Zhao, "Data-driven time-varying inertia estimation of inverter-based resources,", ***IEEE Transactions on Power Systems***, vol. 38, no. 5, pp. 4954-4957, Sept. 2023.

[J7] Maolin Zhu, Hao Liu, Junbo Zhao, **Bendong Tan**, Tianshu Bi and Samson Shenglong Yu, "Dynamic state estimation for DFIG with unknown inputs based on cubature Kalman filter and adaptive interpolation," ***Journal of Modern Power Systems and Clean Energy***, vol. 11, no. 4, pp. 1086-1099, July 2023.

[J8] **Bendong Tan**, Junbo Zhao, *et al.*, "A general decentralized dynamic state estimation with synchronous generator magnetic saturation,", ***IEEE Transactions on Power Systems***, vol. 38, no. 1, pp. 960-963, Jan. 2023.

[J9] **Bendong Tan**, Junbo Zhao, *et al.*, "Distributed frequency divider for power system bus frequency online estimation considering virtual inertia from DFIGs,", ***IEEE Journal on Emerging and Selected Topics in Circuits and Systems***, vol. 12, no. 1, pp. 161-171, 2022.

[J10] **Bendong Tan**, Junbo Zhao, *et al.*, "Decentralized data-driven estimation of generator rotor speed and inertia constant based on adaptive unscented Kalman filter,", ***International Journal of Electrical Power & Energy Systems***, vol.137, pp.107853, 2022.

[J11] **Bendong Tan**, Junbo Zhao, *et al.*, "Power system inertia estimation: Review of methods and the impacts of converter-interfaced generations," ***International Journal of Electrical Power & Energy Systems***, vol.134, pp.107362, 2022. (Best paper award)

[J12] Seyyed Rashid Khazeiynasab, Junbo Zhao, Issa Batarseh and **Bendong Tan**, "Power plant model parameter calibration using conditional variational autoencoder," ***IEEE Transactions on Power Systems***, vol. 37, no. 2, pp. 1642-1652, 2022.

[J13] Juelin Liu, Zhifang Yang, Junbo Zhao, Juan Yu, **Bendong Tan** and Wenyuan Li, "Explicit data-driven small-signal stability constrained optimal power flow," ***IEEE Transactions on Power Systems***, vol. 37, no. 5, pp. 3726-3737, Sept. 2022.

[J14] **Bendong Tan**, Jun Yang, *et al.*, "Spatial-temporal adaptive transient stability assessment for power system under missing data," ***International Journal of Electrical Power & Energy Systems***, vol.123, pp.106237, 2020.

[J15] **Bendong Tan**, Jun Yang, *et al.*, "A deep imbalanced learning framework for transient stability assessment of power system," ***IEEE Access***, vol.7, pp.81759-81769, 2019.

- **Conference Paper**

[C1] **Bendong Tan**, Junbo Zhao, *et al.*, "Time-varying inertia estimation for inverter-based resources,", in ***2023 IEEE Power & Energy Society General Meeting (PESGM)***, pp. 1-5, 2023.

[C2] **Bendong Tan**, Junbo Zhao, *et al.*, "Explainable Bayesian neural network for probabilistic transient stability analysis considering wind energy,", in ***2022 IEEE Power & Energy Society General Meeting (PESGM)***, pp. 1-5, 2022. (Best paper award)

[C3] **Bendong Tan**, Junbo Zhao, *et al.*, "Interpretable data-driven probabilistic power system load margin assessment with uncertain renewable energy and loads,", in ***11th International Conference on Innovative Smart Grid Technologies (Asia)***, 2022. (Best paper award)

[C4] **Bendong Tan**, Junbo Zhao, *et al.*, "Extended frequency divider for bus frequency estimation considering virtual inertia from DFIGs,", in ***2021 IEEE PES Innovative Smart Grid Technologies Conference - Latin America (ISGT Latin America)***, pp. 1-5, 2021.

[C5] **Bendong Tan**, Jun Yang, *et al.*, "A novel temporal feature selection for time-adaptive transient stability assessment,", in ***2019 IEEE PES Innovative Smart Grid Technologies Europe (ISGT-Europe)***, pp. 1-5, 2019.

[C6] **Bendong Tan**, Jun Yang, *et al.*, "Representational learning approach for power system transient stability assessment based on convolutional neural network," in ***The 6th Renewable Power Generation Conference***, Sep. 2017.

- **In Progress Paper**

[J1] **Bendong Tan**, Junbo Zhao, *et al.*, "Variational Bayesian adaptive unscented Kalman filter for time-varying inertia and damping estimation of IBRs with current limiter,", ***IEEE Transactions on Power Systems***, 2023. **(Under review)**

[J2] **Bendong Tan**, Junbo Zhao, *et al.*, "Scalable risk assessment of rare events in power systems with uncertain wind generation and loads,", ***IEEE Transactions on Power Systems***, 2023. **(Under review)**

[J3] **Bendong Tan**, Junbo Zhao, *et al.*, "Bayesian post-fault power system dynamic trajectory prediction,". **(In preparation)**

## AWARDS

- **Conference Participation Award** July 2023
  Awarded by the University of Connecticut
- **GE Fellowship for Excellence** May 2023
  Awarded by the University of Connecticut
- **IEEE Transactions on Power Systems Oustanding Reviewer** Mar. 2023
  Awarded by the IEEE Transactions on Power Systems Editorial Broad
- **Best Conference Paper Award** Aug. 2022
  Awarded by IEEE Conference on Innovative Smart Grid Technologies - Asia (ISGT Asia) for the paper "Interpretable data-driven probabilistic power system load margin assessment with uncertain renewable energy and load"
- **Best Conference Paper Award** July 2022
  Awarded by IEEE Power and Engineering Society General Meeting for the paper "Explainable Bayesian neural network for probabilistic transient stability analysis considering wind energy"
- **Best Journal Paper Award** June 2022
  Awarded by the International Journal of Electrical Power & Energy Systems for the paper "Power system inertia estimation: Review of methods and the impacts of converter-interfaced generations"
- **IEEE Transactions on Power Systems Oustanding Reviewer** Jan. 2022
  Awarded by the IEEE Transactions on Power Systems Editorial Broad
- **Power Exploration Scholarship(top 1%)** Sept. 2019
  Awarded by Wuhan University
- **First-class Undergraduate Scholarship of Wuhan University(top 5%)** Sept. 2016
  Awarded by Wuhan University
- **First-class Undergraduate Scholarship of Wuhan University(top 5%)** Sept. 2015
  Awarded by Wuhan University

## SKILLS

- **Power System Softwares**
  - Skilled with with **DIgSILENT simulation language** for converter-interfaced generation modeling;
  - Proficient in using **PSS/E, PowerWorld** for dynamic power system simulation;
  - Experienced in **PST, PSAT, PSASP, PSD-BPA** for dynamic power system simulation.
- **Programming Languages** – Proficient in using Python and Matlab Julia; Experienced in C, Julia.

## PROJECTS

- **Probabilistic Power System Risk Assessment with Uncertain Renewable Generations and Load**
  *ISO New England* Dec. 2022 – Nov. 2023
  **Major contribution**
  - Expedited assessment of the risk of the ISO New England power system under massive contingencies;
  - Fast risk assessment of forced outages through transfer learning.
- **Load Sculptor: Robust Dynamic Load Modeling and Uncertainty Quantification**
  *US Department of Energy and Lawrence Livermore National Laboratory* June 2020 – Dec. 2023
  **Major contribution**
  - Filter out invalid simulations due to the unsuitable parameter combinations;
  - Global sensitivity analysis is developed to identify the sensitivity of each parameter;
  - Bayesian parameter estimation for the composite load model with distributed generation (CMPLDWG).
- **Modeling and Analytics for WI Near Term Resilience and Reliability**
  *US Department of Energy and National Renewable Energy Laboratory* Mar. 2021– Apr. 2022
  **Major contribution**
  - Uncertainty quantification from uncertain sources, i.e., wind generations and loads, to load margin;
  - Transfer learning-enabled model updating under topology changes.
- **Macro-Resiliency of the North American Power Grid**
  *US Department of Energy and Argonne National Laboratory* June 2020 – Dec. 2023
  **Major contribution**
  - Track the time-varying inertia in inverter-based resources (IBRs) with grid-forming control;
  - Effects from the current limiter are mitigated during inertia tracking.

## TALKS

[T1] Sept. 2023, "Amortized Bayesian parameter estimation approach for WECC composite load model", *2023 UCONN Power/Electronics Joint Seminar*, Srorrs, CT, USA.

[T2] July 2023, "Time-varying inertia estimation for inverter-based resources", *IEEE Power & Energy Society General Meeting (PESGM)*, Orlando, FL, USA.

[T3] Nov. 2022, "Interpretable data-driven probabilistic power system load margin assessment with uncertain renewable energy and loads", *11th International Conference on Innovative Smart Grid Technologies (Asia)*, Online.

[T4] July 2022, "Time-varying inertia estimation for inverter-based resources", *IEEE Power & Energy Society General Meeting (PESGM)*, Denvor, CO, USA.

[T5] Sept. 2021, "Extended frequency divider for bus frequency estimation considering virtual inertia from DFIGs", *IEEE PES Innovative Smart Grid Technologies Conference - Latin America (ISGT Latin America)*, Online.

[T6] Sept. 2019, "A novel temporal feature selection for time-adaptive transient stability assessment", *IEEE PES Innovative Smart Grid Technologies Europe (ISGT-Europe)*, Online.

[T7] Oct. 2017, "Representational learning approach for power system transient stability assessment based on convolutional neural network" *The 6th Renewable Power Generation Conference*, Wuhan, Hubei, China.

## SERVICES

- **Peer Reviewer**
  - IEEE Transactions on Power Systems
  - IEEE Transactions on Smart Grid
  - IEEE Power Engineering Letters
  - IEEE Open Access Journal of Power and Energy
  - International Journal of Electrical Power & Energy Systems
  - CSEE Journal of Power and Energy Systems
  - IET Generation, Transmission & Distribution
  - Journal of Modern Power Systems and Clean Energy
  - IEEE Transactions on Circuits and Systems II: Express Briefs
  - Electric Power Systems Research